

Hematological response and diving response during apnea and apnea with face immersion

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Accepted: 26 April 2007 / Published online: 1 June 2007
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Abstract Increased hematocrit (Hct) attributable to splenic contraction accompanies human apneic diving or apnea with face immersion. Apnea also causes heart rate reduction and peripheral vasoconstriction, i.e., a cardiovascular diving response, which is augmented by face immersion. The aim was to study the role of apnea and facial immersion in the initiation of the hematological response and to relate this to the cardiovascular diving response and its oxygen conservation during repeated apneas. Seven male volunteers performed two series of five apneas of fixed near-maximal duration: one series in air (A) and the other with facial immersion in 10°C water (FIA). Apneas were spaced by 2 min and series by 20 min of rest. Venous blood samples, taken before and after each apnea, were analysed for Hct, hemoglobin concentration (Hb), lactic acid, blood gases and pH. Heart rate, skin capillary blood flow and arterial oxygen saturation were continuously measured non-invasively. A transient increase of Hct and Hb by approximately 4% developed progressively across both series. As no increase of the response resulted with face immersion, we concluded that the apnea, or its consequences, is the major stimulus evoking splenic contraction. An augmented cardiovascular diving response occurred

during FIA compared to A. Arterial oxygen saturation remained higher, venous oxygen stores were more depleted and lactic acid accumulation was higher across the FIA series, indicating oxygen conservation with the more powerful diving response. This study shows that the hematological response is not involved in causing the difference in oxygen saturation between apnea and apnea with face immersion.

Keywords Spleen contraction · Hematocrit · Oxygen conservation · Face immersion · Breath-hold

Introduction

Release of red blood cells into the circulation by splenic contraction may contribute to prolonged diving in seals (Qvist et al. 1986; Hurford et al. 1996). It has been shown that an increase in hematocrit (Hct) and hemoglobin concentration (Hb) is also seen in humans during exercise (Laub et al. 1993; Stewart et al. 2003) and apneic diving (Hurford et al. 1990). Simulated diving by apnea with face immersion, known to evoke the human cardiovascular diving response, also triggers splenic contraction and associated increases in circulating Hct and Hb (Schagatay et al. 2001; Espersen et al. 2002; Bakovic et al. 2003; Schagatay et al. 2005) but it is not clear whether the main stimulus is derived from the apnea or the facial immersion. These Hct and Hb increases are not observed in splenectomized subjects (Schagatay et al. 2001). The human cardiovascular diving response has been extensively studied, and it is known that facial chilling during apnea increases the response, i.e., the vagal reduction of heart rate (HR) and sympathetic reduction of peripheral blood flow (Hurwitz and Furedy 1986). The neural inputs eliciting the human

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cardiovascular diving response are thus both derived from the apnea stimulus, likely mediated by lung stretch receptors, and the cold stimulation of the face, especially the ophthalmic branch of the trigeminal nerve (Schuitema and Holm 1988). Either apnea or facial chilling alone evokes about half the response (Hurwitz and Furedy 1986).

In the present study we evaluated the roles of apnea and facial chilling in the triggering of the hematological response. It was concluded by Espersen et al. (2002) that splenic contraction may also be fortified by face immersion, in a study using apneas and apneas with face immersion spaced by 10–15 min resting intervals. In that study, it was also reported that recovery occurred within a few minutes after apneas, contrary to the observations by Schagatay et al. (2005) where splenic contraction was not reversed until 9 min after serial apneas. It may be motivated to use a serial apnea model with short, e.g. 2 min pauses between apneas, as apneic diving in the field rarely occurs as single dives, but as series of dives with short intervals. Other apneic events, e.g. sleep apnea, often occur in series as well. Therefore, laboratory studies of the physiology of apnea based on single apneic episodes may lead to conclusions not directly applicable to the full adjustments of apneic diving or to the accumulated effects of repeated apneas.

The cardiovascular diving response has been suggested to conserve oxygen during apnea (Schagatay and Andersson 1998; Andersson and Schagatay 1998; Lindholm and Linnarsson 2002) and may provide an important means for sustaining prolonged apneas without the risk of hypoxic syncope. When the diving response is powerful, as during apnea with face immersion, the resulting arterial desaturation is less pronounced than during apnea alone (Andersson and Schagatay 1998; Andersson et al. 2002). This has been ascribed to the effect of both the peripheral vasoconstriction and the bradycardia; i.e., the reservation of the central circulating oxygen supply mainly for the heart and brain and thereby a slower depletion of lung oxygen stores (Andersson et al. 2002, 2004) and reduced oxygen requirements of the cardiac muscle (Lin et al. 1982; Andersson et al. 2004). The cardiovascular response develops fully within 30 s, and while serial diving does not enhance the response (Schagatay et al. 1999), it is fortified by long-term apnea training (Schagatay et al. 2000).

Serial apneas with face immersion evoke a transient increase in Hct and Hb, which is accompanied by a delay in the physiological breaking point of apnea (Schagatay et al. 2001). Neither the hematological effect nor the delay is observed in splenectomized subjects (Schagatay et al. 2001). This has helped to explain the previously poorly understood phenomenon that apneas are prolonged when repeated at short intervals (Hentsch and Ulmer 1984), and has been shown not to be due to increased diving response or hyperventilation (Schagatay et al. 1999). The increase in

Hct and Hb reaches full effect after 3–4 apneas with short intervals, and the physiological breaking point is delayed across the same number of apneas (Schagatay et al. 2001). The total plasma protein levels were not observed to change during the Hct and Hb increase (Schagatay et al. 2001). While this increase was closely linked to the observed volume changes of the spleen, spleen volume changes do not appear to account for all of the observed changes in Hct and Hb (Schagatay et al. 2005). The neural pathways initiating the splenic contraction have not yet been determined.

Adaptive mechanisms, involving both the cardiovascular diving response and the increased blood-gas storage capacity by the blood cell-boosting splenic contraction, may thus serve to facilitate prolonged diving in humans. Repeated diving may elicit these two physiological adjustments at similar or different time frames, by the same or different mechanisms. If the mechanisms for initiation and the time for establishment of the two adjustments are similar or identical, they may be considered as parts of the same general response. We would therefore have to accept a wider definition of the diving response than the classical, i.e., bradycardia and peripheral vasoconstriction. As such, the more recently discovered spleen response may or may not be part of the general cardiovascular diving response, which is a central question in this study. Previously observed differences in arterial saturation after apneas and apneas with facial immersion could, in theory, be due to any of these adaptive adjustments if both are enhanced by facial immersion.

Aim

In the present study the specific aims were to (1) evaluate apnea with or without the presence of facial cold water immersion in the triggering of the human hematological response related to spleen contraction and (2) relate the hematological response to the cardiovascular diving response and its oxygen conservation during repeated apneas. This addresses the question: is the splenic response involved in causing the difference in saturation between apneas with and without face immersion, previously ascribed to the differences in the diving response? A repeated apnea model was used to mimic what happens in most naturally occurring apnea events.

Methods

Subjects

Seven male subjects, aged 22–35 years (mean 27 years), volunteered for the study. The subjects were trained apneic divers from under-water rugby or competitive apnea (5 subjects), or subjects with some breath-hold diving experience

and good breath-holding ability (>2 min; 2 subjects). Their levels of non-diving physical training ranged from 0 to 10 h/week (mean 4.7 h). Subjects' mean (SD) height was 186 (5) cm, weight 78 (10) kg and vital capacity (VC) 6.3 (0.9) L. None of the subjects used nicotine. The protocol was approved by the human research ethics committee at Lund University and complies with the ethics of the Declaration of Helsinki. After receiving a description of the procedures and an explanation of potential risks involved, all subjects gave their written, informed consent.

Procedures

At arrival to the laboratory, after at least 2 h free of food ingestion, caffeine use or exercise, the subjects standing VC was measured and other physical characteristics registered. Subjects then rested in the prone position for at least 30 min before performing apneas. During the resting period a venous catheter was inserted in one arm and probes for registration of other parameters were attached to the opposite hand. The subjects performed two series of five apneas of a fixed near-maximal individual duration, based on separate pre-trials, one series with the face in air (A) and the other with facial immersion in cold water (FIA). Average ambient air temperature (SD) was 22.3 (1.2)°C and water temperature 10.4 (0.7)°C. Subjects were notified when 1 min remained before each apnea, received a noseclip 30 s prior to apnea and started the apnea after 10 s of countdown. Subjects were told to avoid hyperventilation, to exhale and make a deep but not maximal inspiration before apneas, and to keep the chest relaxed without expiring till the end of the apneas after another 10 s of countdown. They received information about the apnea progress every 30 s. Mean (SD) apneic time for A and FIA was 161 (31) s. Apneas of each series were spaced by 2 min resting intervals. The two series were separated by at least 20 min of rest and their order weighted so that every other subject started with FIA.

Measurements

Vital capacity was measured with a spirometer (Spirolite 201, Vise Medical Co. Ltd, Chiba, Japan). Venous blood samples of 2 ml were collected from the catheter 10 min before the test series started, directly before each apnea, directly after each apnea and at intervals of 3, 10 and 20 min after the last apnea. Total blood volume drawn, including the volume of blood drawn to clear the dead space of the catheter and stopcock system, was between 40 and 50 ml per individual. A volume of 15–20 ml of sterile isotonic NaCl solution was infused for rinsing the catheter. Hematocrit, Hb, pH, plasma lactate concentration [p-La]_v, and blood gases were analysed. The anaerobically drawn blood samples were stored on ice water and analysed for

venous oxygen saturation (SvO₂) and pressure (pvO₂), venous carbon dioxide pressure (pvCO₂), Hb and pH within 30 min of sampling (ABL 615, Radiometer, Copenhagen, Denmark). Hematocrit was measured in triplicate by microcentrifugation, and lactate on a lactate analyser (YSI, Yellow Springs, USA). Heart rate and arterial oxygen saturation were recorded using pulse-oxymetry with a finger probe, with the average function set on 6 s (Biox 3700, Ohmeda, Madison WI, USA). Laser-Doppler technique was used for detecting changes in skin capillary blood flow (SkBF) in the thumb (Advance Laser Flowmeter, ALF 21, Advance Co. Ltd, Tokyo, Japan). As the laser-Doppler was not calibrated for absolute flow values, arbitrary laser-Doppler units are presented in the Results section, shown to be reliable for determining relative changes. Continuous parameters were registered from 2 min before till 2 min after each apnea series and stored in a computer together with an event marker.

Data analysis

Subjects served as their own controls. Baseline values used for calculation of changes of the continuous parameters HR, SkBF and arterial oxygen saturation were obtained from the period 90–30 s before the first apneic episode. For discrete measurements, the samples taken before the first apnea or apnea with face immersion were averaged for baseline values, and used as a reference for the apneic values obtained directly after each apnea. The apneic values for HR and SkBF were obtained from the entire apneas except the initial 30 s of decline, as a full development of the diving response requires 30 s (Jung and Stolle 1981). The apneic value used for arterial saturation was the nadir observed during the 60 s period after each apnea. This nadir corresponds to the end of apnea as there is a delay of approximately 20–30 s when finger pulse oximetry is used (Severinghaus and Naifeh 1987). Statistical comparisons were done within series, with pre-apneic reference and between A and FIA series using paired Students *t* test. Significance was accepted at the 5% level.

Results

Effects of facial immersion

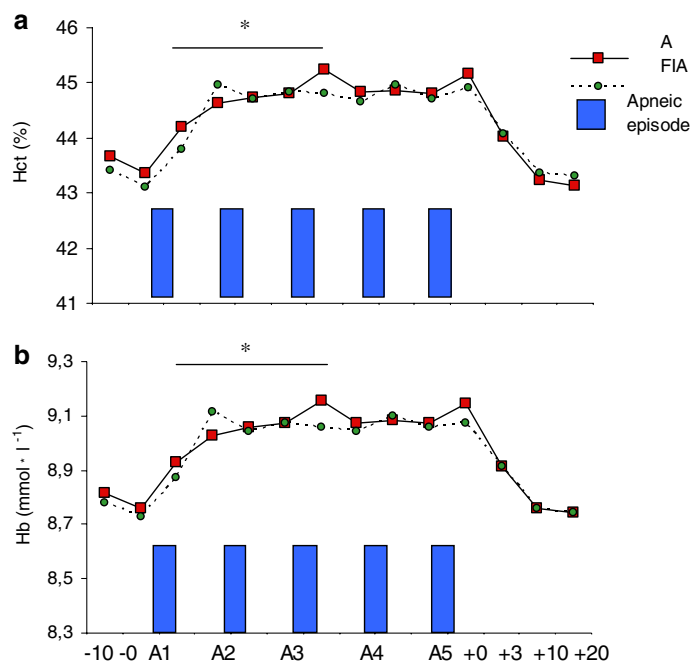
Baseline values between A and FIA before the apnea series were the same for all parameters (NS). The mean of the five apneic values (recorded during or just after each apnea, depending on the parameter) were compared for all parameters with mean baseline, pre apnea-series values for A and FIA, respectively (Table 1). Comparisons between A- and FIA values are shown in Figs. 1–3.

Table 1 Hematological, cardiovascular and metabolic data from apnea and apnea with face immersion

	Apnea (A)			Face immersion apnea (FIA)		
	Baseline before A	Mean from all A	A-Basel. P =	Baseline before FIA	Mean from all FIA	FIA-Basel. P =
Hct (%)	43.5 (1.3)	44.8 (1.1)	0.007	43.3 (0.9)	44.6 (1.1)	0.006
Hb (mmol l ⁻¹)	8.8 (0.3)	9.0 (0.2)	0.004	8.8 (0.2)	9.0 (0.2)	0.010
HR (beats min ⁻¹)	74.4 (6.2)	66.3 (3.9)	0.045	76.4 (5.7)	58.3 (3.2)	0.003
SkBF (arbitrary units)	9.5 (3.7)	8.6 (2.7)	NS	9.3 (3.4)	6.4 (2.6)	0.064
SaO ₂ (%)	98.3 (0.4)	87.5 (1.8)	0.000	98.2 (0.4)	90.5 (1.3)	0.000
SvO ₂ (%)	74.6 (5.6)	62.6 (5.1)	0.005	75.5 (4.8)	58.1 (3.2)	0.001
pvO ₂ (mmHg)	39.2 (3.4)	33.1 (2.5)	0.033	43.1 (2.7)	31.1 (0.6)	0.002
pvCO ₂ (mmHg)	46.0 (1.4)	47.5 (1.4)	0.082	45.7 (1.1)	47.2 (0.9)	0.044
[p-La] _v (mmol l ⁻¹)	0.56 (0.08)	0.63 (0.06)	0.053	0.54 (0.06)	0.64 (0.08)	0.024
pH	7.380 (0.005)	7.376 (0.006)	NS	7.381 (0.010)	7.376 (0.006)	NS

Means (SE) of baseline values and apnea values for seven subjects. Apnea values for heart rate (HR) and SkBF were obtained during each apnea and for hematocrit (Hct) hemoglobin concentration (Hb), arterial oxygen saturation (SaO₂), venous oxygen saturation (SvO₂) and pressure (pvO₂), venous carbon dioxide pressure (pvCO₂), venous lactate concentration [p-La]_v and pH values were obtained directly after each apnea. *P*-values represent paired *t* tests of mean apnea values compared to baseline values, for A- and FIA-series, respectively. *P*-values >0.1 not indicated

Fig. 1 Hematocrit (Hct) and hemoglobin concentration (Hb) before and after each of five apneas (A) or apneas with face immersion (FIA) of the same, fixed duration, and during post-series recovery. Means for seven subjects are shown. Standard error is left out for clarity. When A and FIA values were compared no differences were observed between the series or between individual data points. In both A and FIA series, Hct and Hb had increased more after A3 than after A1 (* indicates *P* < 0.05) whereas no further increase occurred after A3



Hematological response

A transient increase in Hct and Hb, by approximately 4% after the fifth apnea, was found across both apnea series irrespective of the cold stimulus (Fig. 1). The mean post-apneic values from all apneas reveal lower increases (Table 1), but as can be seen in Fig. 1 the response developed gradually across A1–A3 and remained at this level till the fifth apnea. During the A-series, the mean (SE) Hct increased from 43.4 (1.3)%, just before the first apnea, to 45.2 (1.1)% after the fifth apnea (by 4.2%; *P* < 0.05; Fig. 1),

and had returned to baseline after 10 min. During the FIA-series, Hct increased from 43.1 (0.6) to 44.9 (1.0)% after the fifth apnea (by 4.2%; *P* < 0.05). The Hb value increased during the A-series from 8.76 (0.28) mmol/l, before the apneas, to 9.14 (0.23) mmol/l after the fifth apnea (by 4.4%; *P* < 0.01), and during the FIA-series it increased from 8.73 (0.14) to 9.07 (0.22) mmol/l (by 3.9 %; *P* = 0.05). Recovery values were completely restored to baseline levels, 10 min after the end of apneas in both series (Fig. 1). Values of Hct and Hb did not differ between A and FIA for the pooled values from all apneas or at any point between series (Fig. 1).

In both A and FIA series, the Hct and Hb had increased more, after A3 than after A1 ($P < 0.05$) whereas no further increase occurred after A3 (NS).

Cardiovascular diving response

Heart rate was transiently reduced during all apneas, but returned to pre-apneic baseline levels during the intervals between apneas (Fig. 2a). The magnitude of the response was the same for each apnea across the series but clearly different between the series. Skin blood flow tended to be reduced during apneas, and increased beyond baseline during recovery (Fig. 2b). During FIA the HR fell from 76 bpm to 58 beats per minute (by 24%; $P < 0.01$), and SkBF tended to be reduced from 9.3 to 6.4 laser-Doppler units (by 31%; $P = 0.06$; Table 1). During A, HR fell from 74 to 66 beats per minute (by 11%; $P < 0.05$) and SkBF changed marginally from 9.5 to 8.6 units (9%; NS; Table 1). After apneas in the A series, the SkBF recovery value increased beyond baseline and the values after A4 and A5 were higher than that after A1 ($P < 0.05$). Values of both HR and SkBF differed between A- and FIA-series, both for the pooled values from all apneas (both $P < 0.001$) and for individual data points (Fig. 2), with FIA evoking the strongest diving response.

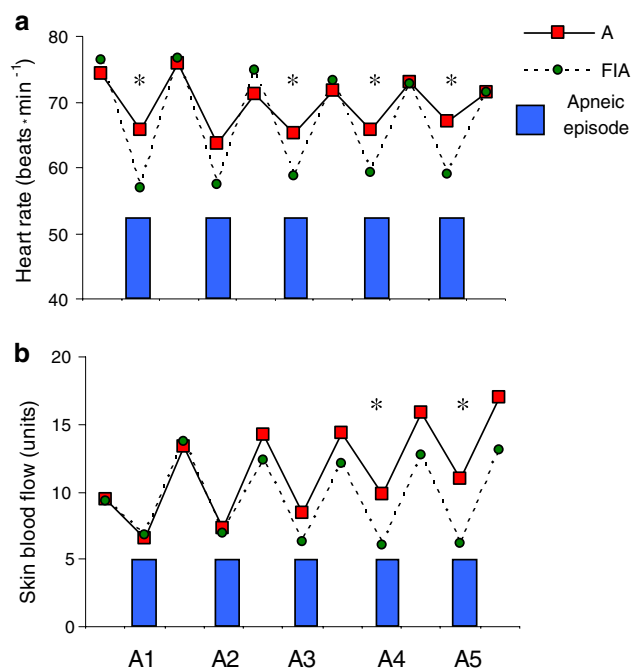


Fig. 2 Heart rate (HR) and skin blood flow (SkBF) before, during and after each of five apneas (A) or apneas with face immersion (FIA) of the same, fixed duration. Means for seven subjects are shown, with standard error left out for clarity. Difference between A and FIA values are indicated by * for $P < 0.05$

Metabolic parameters

Arterial oxygen saturation (SaO_2) decreased as a response to both A and FIA but the magnitude of decrease was greater during A, with a mean reduction of 11.0% from baseline compared to 7.8% during FIA (both $P < 0.001$; Table 1). When the pooled values were compared between the A- and FIA-series, there was a more pronounced arterial desaturation after A ($P < 0.05$). The difference in SaO_2 between A and FIA series was also evident when values from single apneas were compared (Fig. 3a). During both A and FIA, there were reductions from baseline of venous oxygen saturation (SvO_2) and pressure (pvO_2) with the most pronounced reductions during FIA (Table 1). During A, SvO_2 was reduced by 16.1% and during FIA by 23.1% (both $P < 0.01$). During A, pvO_2 was reduced by 15.4% ($P < 0.05$), compared to 27.9% during FIA ($P < 0.01$; Table 1). When the pooled values were compared between the A and FIA-series, there was a more pronounced venous desaturation after FIA ($P < 0.05$). Differences in SvO_2 and PvO_2 between A and FIA were also evident towards the end of the series when values from single apneas were compared (Fig. 3b–c). The venous carbon dioxide pressure (pvCO_2) increased from baseline only during FIA ($P < 0.05$; Table 1). However, no differences were observed when the pooled values or individual data points were compared between A- and FIA-series (Fig. 3d).

The $[\text{p-La}]_v$ tended to rise above baseline in A (by 12.5%; $P = 0.053$) and in FIA it rose by 17.2% ($P < 0.05$; Table 1), but pH showed no changes from baseline after apneas. Neither $[\text{p-La}]_v$ nor pH differed significantly between absolute values for A- and FIA-series (Fig. 3e–f).

Discussion

Effects of facial immersion

This study showed that the hematological response, unlike the cardiovascular diving response, was evoked to the same extent during apneas as during apneas with face immersion. The rises in Hct and Hb during apnea have been linked to splenic contraction by other studies (Schagatay et al. 2001; Espersen et al. 2002; Bakovic et al. 2003; Schagatay et al. 2005). This indicates that splenic contraction occurred to the same extent during apnea irrespective of face immersion, suggesting that apnea or any of its consequences is the major stimulus responsible for its elicitation. This is contrary to the findings by Espersen et al. (2002), reporting that apnea with face immersion triggered a more powerful splenic contraction than apnea alone. A difference between their study and the present one is, their use of apneas separated by long pauses which would likely have lead to the development of a less

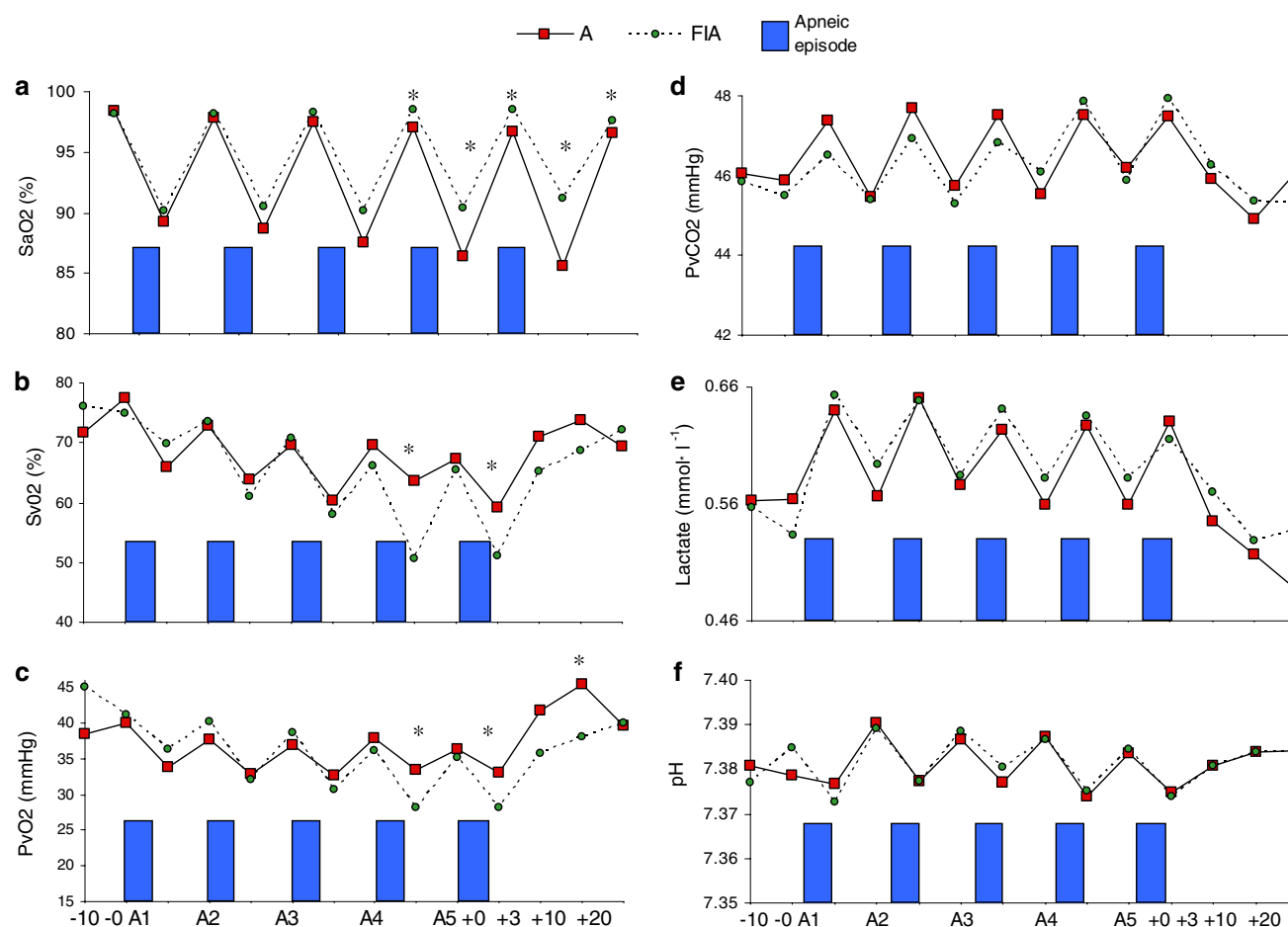


Fig. 3 Mean arterial and venous hemoglobin saturation (SaO₂; SvO₂), venous oxygen and carbon dioxide pressure (PvO₂, PvCO₂), lactate concentration ([p-La]_v) and pH before and after each of five apneas (A)

or apneas with face immersion (FIA) of a fixed duration. Means for seven subjects are shown. Standard error is left out for clarity. Difference between A- and FIA values is indicated by * for $P < 0.05$

than maximal response, compared to our study. The neural input from the apnea could be derived from the apnea itself, by the cessation of cyclic stimulation of lung stretch receptors or by direct signalling from the respiratory (apneustic) centre, or it could be related to chemoreceptor input by its effects on arterial O₂ and/ or CO₂. In the latter case, a more pronounced arterial desaturation during A, compared to FIA, could possibly trigger a stronger splenic contraction, and any effect of the cold stimulation on the face could be a contributing factor, balancing the responses between A and FIA. But early in our apnea series the SaO₂ value is the same between A and FIA, yet we see no differences between Hct and Hb responses. It remains to be seen if chemoreceptor input is involved in evoking or modifying the response and, if so, to what extent the SaO₂ difference present in our study could have such an effect.

Development of the hematological response

The findings of the present study show that the hematological response develops progressively across 3–4 apneas,

while the bradycardia of the diving response reaches its full magnitude during each apnea, and is fully reversed between apneas. Several apneas were needed for a full hematological response to develop, and the 2 min pauses were too short to allow recovery. In fact, the Hct and Hb continued to rise in the intervals between apneas 1–3. This slower onset of the hematological response may be due in part to a circulatory delay. Circulation time may thus prolong the increase of the number of circulating red blood cells beyond the time for re-expansion of the spleen. However, in a study with a similar protocol we used Ultrasonic imaging for spleen volume measurements, and recorded a re-expansion of the spleen over 8–9 min (Schagatay et al. 2005). In the present study, full recovery of Hct and Hb after five apneic episodes occurred within 10 min, which is also the time needed to reverse the “short-term training effect”, i.e., the increase in apneic duration seen during repeated maximal effort apneas (Schagatay et al. 1999). Espersen et al. (2002) reported a recovery after 2 min, but Bakovic et al. (2003) reported that only partial recovery of

spleen volume occurred within 60 min after serial apneas. The present study therefore suggests an intermediate recovery time. This apparently slower onset and recovery of the hematological response compared to the cardiovascular diving response, supports the view that the mechanisms for evoking the responses may be different. It also suggests that the hematological response could have an important role in the recovery situation between apneas. If circulating catecholamines are involved in initiating the spleen response during apneas just as they have been suggested to be during exercise (Stewart et al. 2003), it is likely that these levels remain high between apneas, but may be reduced when an apnea series has been completed. Thus, the recovery situation could be different depending on whether the apnea is directly followed by another one or not, which could possibly be a contributing explanation for the differences in the responses between the apneas in the present study and the study by Espersen et al. (2005). However, after our apnea series is completed, the recovery time is still beyond that found by Espersen et al. (2002), which may be due to an enhanced response by repeated apneas in our study.

The identical hematological response during FIA and A in the present study, in combination with the more clearly pronounced cardiovascular diving response during FIA, indicates that the initiation of the hematological response and the classical diving response are different, which suggests that these adjustments may in fact represent different responses. This is further supported by the clear difference in response time and recovery time of the hematological and the diving response. Thus, while the cardiovascular diving response involves concurrent sympathetic and parasympathetic activation (Hurwitz and Furedy 1986), the splenic contraction may possibly be related to a humoral response, i.e., increased levels of plasma catecholamines (Stewart et al. 2003) or a combination of neural and humoral factors.

Oxygen conservation during serial apneas

When all values were compared between series, the difference between A and FIA for HR, SkBF, SaO_2 , SvO_2 and PvO_2 , summarily indicates that the arterial oxygen stores are conserved at the expense of the venous stores. In addition, the anaerobic metabolism was somewhat increased during FIA, evident from the increased [p-La] when the diving response is more powerful. Such an increase of [p-La] was not present during A with a weaker diving response. The fact that Hct and Hb responses are the same between series, shows that the cause of the metabolic differences is not differences in these parameters.

The finding that the diving response (HR and SkBF reductions) was more powerful during apnea with facial immersion is in accordance with earlier studies (Hurwitz

and Furedy 1986; Schuitema and Holm 1988; Marsh et al. 1995). Less arterial desaturation associated with a more powerful diving response, when apnea is combined with facial cold water immersion, has been found previously during rest (Andersson and Schagatay 1998) and at exercise (Andersson et al. 2002; 2004), leading to the conclusion that the diving response reduces oxygen consumption during apnea. It appears that when a powerful diving response occurs during FIA, involving dramatic peripheral vasoconstriction and bradycardia, anaerobic metabolism increases and the tissue- and venous oxygen pools are depleted to a larger extent than during A. The preserved arterial and lung oxygen stores during FIA will make the oxygen supply to the brain last longer and will delay the point of syncope occurrence during prolonged diving. When the subjects are not breathing the arterial oxygen is used up more slowly mainly by the vital organs, and subsequently there is a slower depletion of lung oxygen, when there is a powerful diving response.

The present results confirm that it is the cardiovascular diving response, rather than the hematological response, which is responsible for the difference in oxygen conservation between apnea and apnea with face immersion. The increased hematocrit may prolong apneic duration across the first 3 apneas in the series, but to the same degree irrespective of face immersion or not. During the short intervals between apneas the reversal of the cardiovascular diving response but maintained spleen response will facilitate recovery by increasing peripheral circulation and at the same time provide a maximized supply of circulating erythrocytes for facilitation of gas transport. If the cardiovascular diving response was not fully reversed between apneas, or if the hematological response was reversed, recovery from apnea would be delayed.

Several parameters showed a progressive change throughout the series, with differences developing between A and FIA after three or four apneas. This was likely due to an incomplete recovery of some parameters in the 2 min pauses between apneas. The situation appears to resemble that during repeated diving, or sleep apnea, with a progressive development of arterial desaturation, and demonstrates that serial apneas may mimic the situation in naturally occurring apneic events.

Conclusions

Fixed near-maximal duration apneas will induce an increased amount of circulating red blood cells via splenic contraction, as previously observed after maximal duration apneas (e.g. Schagatay et al. 2001). Unlike the cardiovascular diving response, the hematological response is not fortified by facial cold-water immersion, but appears to be

triggered by the apnea stimulus alone. While the diving bradycardia reached the same magnitude during each subsequent apnea, the hematological response developed progressively across several apneas and its effect was not fully reversed until 10 min after the apnea series. The hematological response and the diving response may thus be closely associated but not neurally linked, suggesting that they could be viewed as two separate responses. This study further shows that different magnitudes of the cardiovascular diving response, not the hematological response, cause the difference in oxygen saturation after apnea and apnea with face immersion.

Acknowledgments We thank all the subjects for participating in the experiments and the laboratory personnel at the August Krogh Institute for assistance with measurements and blood analysis. The study was supported by grants from the Swedish National Centre for Research in Sports (CIF) and the Swedish defence research establishment (FOA).

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